

Supplementary Winter Examination – 2023

Course: B. Tech. Branch : Civil Engineering Semester : 8th

Subject Code & Name: Energy Efficiency Acoustics and Day lighting in Building

BTCESS802A

Max Marks: 60

Date:17-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) What are the environmental factors and their effects on building?	CO 1	6
B) Explain the factors affecting on human comfort?	CO 2	6
C) Write a short note on lighting principles and its fundamentals.	CO 1	6
Q.2 Solve Any Two of the following.		12
A) Explain the problems related with tropical environment.	CO 1	6
B) Explain in detail human response to thermal environment		6
C) Explain in brief tropical environment.	CO 3	6
Q. 3 Solve Any Two of the following.		12
A) Write a short note on Tropical environments and site environments.	CO 3	6
B) What are the types of heat transfer in buildings?	CO 4	6
C) Illustrate Heat Flow in buildings.	CO 4	6
Q.4 Solve Any Two of the following.		12
A) What is Admittance Method?	CO 5	6
B) Explain Noise pollution in detail	CO 1	6
C) Explain process of heat exchange of building with environment.	CO 4	6
Q. 5 Solve Any Two of the following.		12
A) How the heat transferred in building	CO 4	6
B) Explain the limitations of natural ventilation design	CO 2	6
C) State & explain effect of solar radiation on building.	CO 3	6

*** End ***